

ABHILASHA TANDON

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EDUCATION

Master of Science in Computer Engineering, May 2026

University of Central Florida — Orlando, FL

- **Relevant Coursework:** Operating Systems Design Principles, Software Engineering, Advanced Computer Architecture

Bachelor of Arts in Computer Science and Mathematics, May 2024, GPA: 3.6

New York University — New York, NY

- **Relevant Coursework:** Fundamentals of Machine Learning, Natural Language Processing, Operating Systems, Computer Systems Organization, Numerical Computing, Computer Graphics, Computer Vision

EXPERIENCE

Aegis Medical Group | Leesburg, FL

Fullstack Engineer | September 2024 – December 2025

- Leveraged web development expertise to design and implement React-based frontend components and integrate Strapi as a headless CMS, empowering non-technical stakeholders to manage content independently while ensuring a maintainable and scalable codebase
- Implemented security controls by building and managing REST APIs with MuleSoft to ensure secure communication between Salesforce and EClinicalWorks (ECW), reducing manual updates and streamlining clinical workflows
- Collaborated with cross-functional teams in an agile environment and contributed to sprint planning, standups, and code reviews, demonstrating proficiency in agile development to ensure engineering initiatives aligned with clinical operations requirements
- Implemented containerization and system configuration strategies to automate key processes, resulting in decreased latency, increased system reliability, and simplified workflows enhanced operational efficiency and service quality.
- Created and supervised scripts to automate key processes, reducing latency by 31%, increasing system reliability, and streamlining workflows, demonstrating effective use of foundational knowledge of CI/CD pipelines in enhancing operational efficiency and service quality

TECHNICAL SKILLS

Programming Languages: C, C++, JavaScript, Python, Rust, TypeScript

Frontend: React, HTML/CSS, Tailwind, React.js, Next.js

Backend & APIs: MuleSoft, REST APIs, Strapi, Node.js

Machine Learning & AI: TensorFlow, PyTorch, Scikit-Learn, NumPy, Pandas, PostgreSQL

Software Engineering: Git/GitHub, VS Code, Anaconda, npm, pip

PROJECT LIST

CASSiopeia — Rust

- Created and built a computer algebra system from scratch in Rust, implementing a full expression parser and evaluator spanning over 6,000 lines of code, implementing AST parsing and arbitrary precision arithmetic
- Engineered custom recursive descent parsing logic to handle complex mathematical expressions, applying data structures and algorithmic techniques such as expression trees and operator precedence climbing
- Architected the system with clean module boundaries and well-documented interfaces, enabling iterative feature additions while maintaining correctness and performance

Spotify Genre Classifier — Python, REST API, scikit-learn

- Leveraged the Spotify RESTful API to retrieve and process playlist and listener behavior data for over 300,000 songs, building a robust data pipeline with clean error handling and reproducible results
- Applied advanced analytical techniques including TF-IDF, cosine similarity, UMAP, and spectral clustering to classify 6,000+ artists by genre and surface niche micro-genres
- Automated end-to-end pipeline from data ingestion through model evaluation documenting design decisions and usage so system could be extended or audited by other engineers